

Original Architecture Component	Target Architecture Component	Migration Action & Future-Proofing Strategy
Kafka <i>(On-Premise Event Streaming)</i>	Retained & Redirected Kafka Edge Streams	Repurpose: Instead of streaming events into intermediary, proprietary on-prem systems or virtualization sinks, Kafka is re-routed to stream real-time data payloads straight into Cloud Object Storage buckets using open table standards.
SFTP & Custom APIs <i>(Batch Transfer Mechanisms)</i>	Cloud-Native Object Ingestion	Modernize: Batch exchange frameworks are modified to land raw files directly inside secure cloud storage landing zones. Custom scheduling code is minimized by utilizing localized, lightweight orchestration tools.
Denodo (SeDAAP) <i>(Logical Data Virtualization & View Blending Engine)</i>	Decoupled Cloud Compute (Snowflake or Databricks) + dbt Transformations	Decommission/Downscope: Years of legacy virtualized views, multi-silo data blending, and historical query logic are extracted out of Denodo. These transformations are entirely rewritten as standard ANSI SQL models managed by dbt (Data Build Tool) to isolate logic from vendor lock-in. Denodo is phased out or limited strictly to legacy on-prem connectivity.
Denodo Embedded Catalog <i>(Metadata Tracking & Schemas)</i>	Independent Open Data Catalog (Apache Polaris / Unity Catalog Open Source)	Replace: Metadata discovery, table tracking, and schema states are moved away from Denodo's boundary and into an engine-neutral, open-source centralized metastore catalog. This ensures that changing or adding a query engine later is a basic configuration update rather than a pipeline re-write.
Custom Approval Workflow Module <i>(Access Requests to Datasets)</i>	Decoupled DataGovOps Control Plane	Evolve Asset: The existing approval logic is saved but completely disconnected from Denodo's proprietary catalog layer. It is configured to display data assets from the new Open Data Catalog . When access is approved, it outputs declarative Policy-as-Code variables.
Manual Authorization / View Grants <i>(Security Enforcer Layer)</i>	Embedded & Invisible Runtime Execution	Automate: Instead of a database administrator or developer manually translating permissions into Denodo views, the declarative YAML/JSON files generated by your approval plane are automatically parsed by the target cloud platform. Fine-grained row-filtering and column-level PII masking are triggered seamlessly at query runtime with zero human friction.

DataConnect <i>(Consumer Serving Layer & Endpoints)</i>	Engine-Agnostic Open Serving Layers	Replatform: Consumers, analytical applications, and business intelligence dashboards bypass the legacy virtualization bottlenecks and execute performant, concurrent queries natively against open file structures via your chosen cloud platform compute endpoints.
On-Premise Physical Storage Silos <i>(Underlying Storage Infrastructure)</i>	Open Cloud Storage Planes (AWS S3 / Azure Blob via GCC+)	Migrate: Core physical data residency is centralized within secure government commercial cloud storage buckets. Data files are strictly formatted into standardized, open formats— Apache Iceberg or Delta Lake —retaining complete physical data ownership.